

KANCHAN-AI

Multi-Modal Spurious Gold Detection



Kanchan (कांचन) - Sanskrit & Hindi for pure, refined gold.

Every standard gold-purity test reads the surface. We built the first branch-deployable system that catches what's underneath - by reading bulk material, not just the surface

₹15.6 lakh crore: India's gold loan market (Nov 2025)

9 crore active loan packets nationally

₹0 cost to detect tungsten with any existing branch tool

SuRaksha Cyber Hackathon 2.0 • Grand Finale •

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The Problem: A Penetration-Depth Blind Spot

Real-World Stakes

India's gold-loan exposure stands at **₹15.6 lakh crore** (Nov 2025) across ~9 crore active loan packets.

(Source: [AngelOne](#))

A structural blind spot at this scale is a **systemic exposure**, not a rare edge case.

When gold is pledged as loan collateral, the bank must verify it isn't spurious. The industry's three standard tools - touchstone, XRF spectroscopy, and visual hallmark inspection - all share one structural weakness: **they only interrogate the outer few micrometres of the item.**

XRF Penetration Depth

10 - 15 μm - a genuine-gold shell up to 200 μm defeats it entirely, also it has a very high cost of operation & procurement

Touchstone Read Depth

1-2 μm surface streak only. The acid reaction is the actual ground truth, we do not have acid at branch level, so this is a visual consistency check only.

Tungsten Density Trap

Tungsten: 19.25 g/cm³. Gold: 19.30 g/cm³. Difference: 0.26%, inside typical field measurement uncertainty of $\pm 1-3\%$. Archimedes alone cannot resolve this.



The Physical Insight: Stiffness Closes the Gap Density Can't

No known metal matches gold's density and its stiffness simultaneously.

Density (ρ)

Gold 19.30 g/cm³ vs. Tungsten 19.25 g/cm³

Ratio: **0.997×** - indistinguishable in the field

Young's Modulus (E)

Gold 79 GPa vs. Tungsten 411 GPa

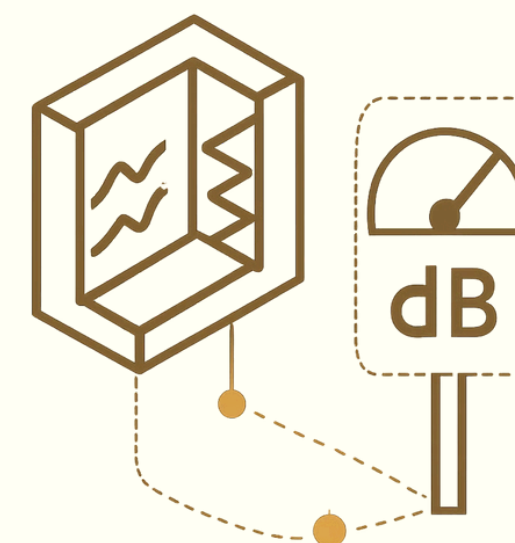
Ratio: **5.2×** - a categorical, order-of-magnitude-adjacent gap

Acoustic Velocity

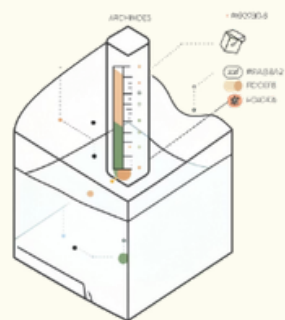
$v = \sqrt{E / \rho}$ - since E differs by 5.2× while ρ differs by only 0.3%, acoustic velocities diverge sharply

Gold rings at approximately 3,240 m/s. Tungsten rings at approximately 5,170 m/s. The 59% velocity gap is 20× larger than measurement noise.

- ❏ A genuine gold shell does not change the stiffness of the core beneath it. Acoustic velocity is a bulk property, it travels through the whole item, not just the surface. A tungsten core with a 200μm gold shell rings like tungsten regardless of what the outside looks like.

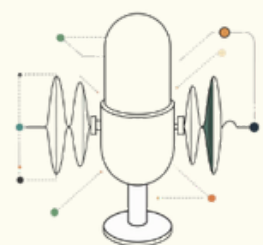


This is the physical rationale for why an acoustic test is the correct complement to density. The two signals are independent, physically motivated, and together cover what neither can cover alone.



Density (Archimedes)

Weigh it in air. Weigh it in water.
Real gold of a given karat has a predictable density. Anything that does not match gets flagged.



Acoustic (Tap-Ring Analysis)

Tap it. Record the ring. Gold has a characteristic sound. Tungsten — the hardest fake to catch — sounds measurably different despite weighing almost the same. This is the test no existing branch tool runs.

Five Physical & Computational Tests - Each Independent, Each Verifiable



Visual (Fiducial-Calibrated CV)

Photograph it with a reference card. The system checks the colour, looks for tarnish, and counts the gemstones. Stones are deducted from the weight before the loan is calculated.



Stone Detection (Decision-Relevant)

Stones are not gold. RBI requires them deducted before the loan is calculated. System finds every stone, estimates its weight, subtracts it automatically. Officer does not do this manually.



3D Reconstruction (Visual / Explanatory)

Same photos, different job. Builds a rotatable 3D model for the case file. Not used in the verdict. Visual record in case the customer later disputes the returned item.

📌 No single test is claimed sufficient in isolation. The system's real strength is what happens when the tests disagree.

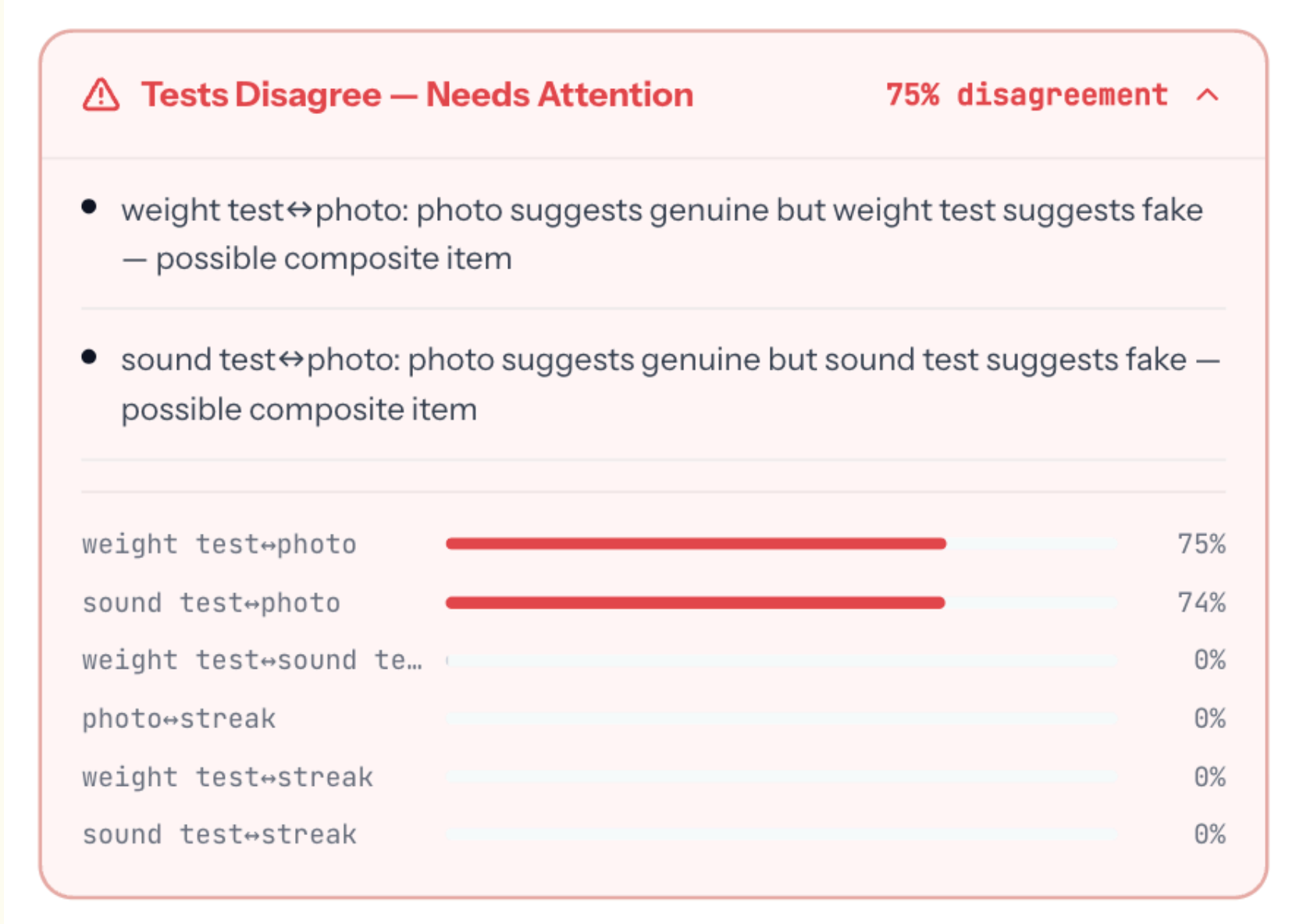
Novelty: Cross-Modal Contradiction as an Explicit Fraud Feature

Every published multimodal fraud-detection system fuses signals expected to agree, using disagreement only as noise to average out. **We inverted that assumption.** All **six pairwise disagreement scores** across four modalities are computed as **first-class fraud features**, not correction terms:

- Density ↔ Acoustic
- Density ↔ Visual
- Density ↔ Streak
- Acoustic ↔ Visual
- Acoustic ↔ Streak
- Visual ↔ Streak

The tungsten signature: density reads gold-consistent (within 0.26% noise floor) while acoustic ring-pitch reads anomalously stiff, neither signal alone is damning; **the conjunction is the fingerprint.**

"No published multimodal fraud-detection architecture we found treats inter-modal contradiction itself as a scored feature. We searched, and we're happy to be corrected."



System Architecture: From Raw Signal to Explainable Verdict

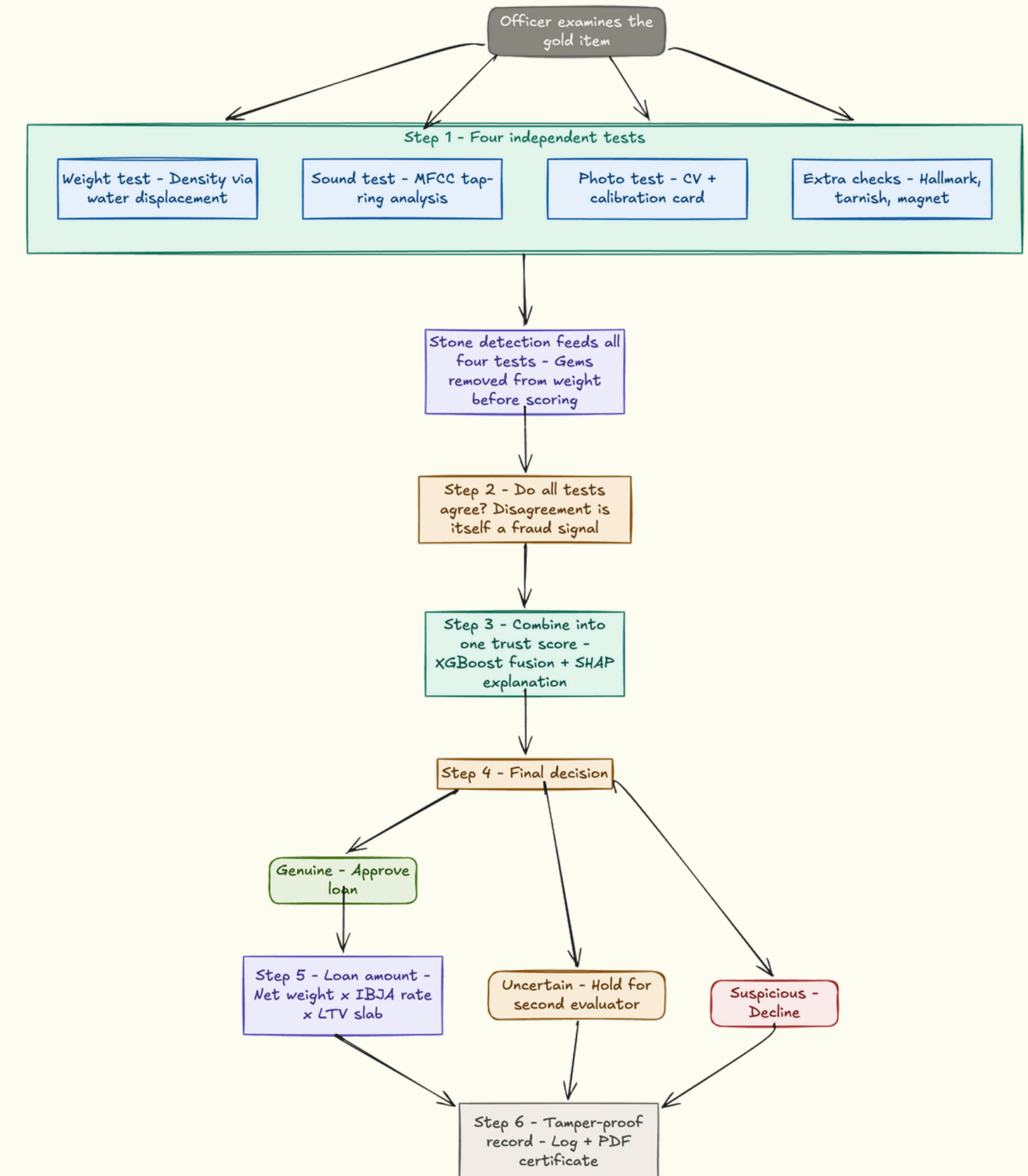
Two Hard Overrides

- **density_risk ≥ 0.85** → REJECT (final, cannot be altered downstream)
- **Ring-pitch stiff-core flag** → REJECT (gold-consistent density + tungsten-consistent stiffness)

Explainability Layer

SHAP TreeExplainer per-feature attribution (<3ms) → Groq Llama 3.3-70B → Gemini 1.5 Flash → deterministic heuristic fallback.

📄 *The LLM explains the verdict. It never decides it. If every LLM provider is unreachable, the officer still gets a complete, auditable verdict.*



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How This Result Was Reached — Step by Step

8 steps · 2 flagged

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Weight-in-water measurement

$\rho = 12.9616 \pm 0.5188 \text{ g/cm}^3$, $V = 0.1705 \text{ cm}^3$

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Does the density match the declared karat?

Within the expected range; chance it matches: 42.3%. Physics best match: no known gold alloy or common metal Note: measurement precision is limited for an item this light — the reading passes, but a heavier sample or repeat weighing would be more conclusive.

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risk = P(true ρ outside karat band | measurement), Gaussian CDF

declared band (g/cm³)

12.8 – 13.4

raw density risk

0.5768

closest fake metal

–

tungsten blind spot

false

measurement adequate

false

Source: Band derived from IS 1417:2016 fineness + CRC densities (mixture rule); scoring per JCGM 106

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Separating the item from the background

Backdrop could not be separated — the photo is set aside (no visual evidence used); retake on a plain background

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Sound test (ring of the tapped item)

risk 0.5

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Checking the ring pitch against genuine gold

Ring pitch 5446 Hz is inside the calibrated genuine band (5533–6997 Hz)

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Sound speed $v = \sqrt{\text{stiffness } E / \text{density } \rho}$ — a stiffer core rings higher. Pitch is compared to a CALIBRATED genuine band, and cross-checked against the length-adaptive expectation $v = 2 \cdot L \cdot f$ when the calibration card gives item size.

measured pitch (Hz)

5446.4

genuine band (Hz)

5533.1 – 6997.1

recording SNR (dB)

67.6

SNR before / after cleanup (dB)

48.2 → 67.3

noise cleanup

on

Source: E per CRC/ASM (Au 79 GPa, W 411 GPa); band via scripts/calibrate_acoustic.py

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Combining all tests & checking for disagreement

combined risk 0.5768, biggest disagreement 0.0768

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Final decision

BORDERLINE (LOW) → HOLD

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Net gold weight, assessed value & RBI-tiered LTV

Net gold 2.21 g (gross_weight). ₹8,546/g at 14K → value ₹18,888; LTV 85% (up to ₹2,50,000) → max loan ₹16,054

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Fusion Engine: How the Verdict Actually Gets Made

- 1

Default Engine

Reliability-weighted log-odds combination, not a black-box classifier. XGBoost available as opt-in alternative.
- 2

Externalized Policy Thresholds

Tunable via decision_policy.json by the bank - no model code changes needed: genuine_high <0.25 · genuine_medium <0.45 · borderline <0.75 · density_override at 0.85
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Hard Override Logic

density_risk ≥ 0.85 → auto REJECT; ring-pitch stiff-core flag → auto REJECT. Both implemented and enforced in code.
- 4

Enforced Safeguard

System will not return APPROVE on an incomplete case - missing weight, photo, or audio automatically downgrades to BORDERLINE. Implemented and enforced in code today.

Demonstration



Results & Cost Comparison vs. XRF

We are not competing with XRF. We cover its specific blind spot.

Dimension	XRF	KANCHAN-AI
Penetration Depth	10–15 µm only	Surfaces subsurface contradiction patterns
Tungsten-Core Detection	Fails (surface-only)	Ring-pitch ratio + contradiction override
Hardware Cost	₹2–5 lakh/unit	~₹13,000 total kit
Branch Deployability	~200 branches nationally	All 9,700+ branches (CPU-only)
Explainability	Basic instrument report	SHAP + plain-English LLM layer

5.2×
Stiffness Gap
Young's Modulus difference, the
physical foundation of acoustic
detection

9700+
Branches Reachable
vs. ~200 with XRF - CPU-only, no
GPU required

Potential Benefits of KANCHAN

Not a research prototype. A system built for the branch desk, the audit trail, and the customer standing across the counter.

Faster Appraisals

Peak season backlogs cut from 35 minutes to under 5. Weight auto-fed. Certificate auto-generated. No documentation gaps.

Customer Disputes Resolved

Timestamped photo at acceptance and at return. Hash-verified. If a customer claims damage or swapping - the proof is already in the file.

3D Model as Baseline

Discolouration, rust, hallmark tampering - visible against the acceptance model at return. The counterfeiter cannot fake a baseline they never saw.

- ❏ Every fraud case in the real world had one thing in common - no one could prove what the item looked like when it walked in. We fix that and detect spurious gold with research backed approach.